NASSAU CANDY DISTRIBUTOR

Factory Reallocation & Shipping Optimization

Recommendation System

A Data-Driven Approach to Supply Chain Decision Intelligence

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Abstract

Nassau Candy Distributor, a leading confectionery wholesale operation, manages 15 product SKUs across five geographically dispersed manufacturing facilities spanning Arizona, Georgia, Minnesota, Illinois, and Tennessee. The current factory-to-product assignment relies on static legacy rules, resulting in suboptimal shipping distances, elevated lead times in certain customer regions, and margin erosion driven by logistics inefficiencies.

This research paper presents a comprehensive Factory Reallocation and Shipping Optimization Recommendation System — a machine learning-powered decision intelligence framework that transcends descriptive analytics. The system employs Linear Regression, Random Forest Regression, and Gradient Boosting models to predict shipping lead times under any factory-product-region configuration. A scenario simulation engine evaluates all alternative factory assignments, ranking recommendations by lead-time reduction, profit impact, and confidence score. The resulting Streamlit dashboard enables real-time what-if analysis and operational decision support for Nassau Candy's leadership team.

Keywords: Supply Chain Optimization, Machine Learning, Predictive Modelling, Factory Reallocation, Shipping Lead Time, Gradient Boosting, Decision Intelligence

SECTION 1

Introduction

The global confectionery distribution landscape is increasingly defined by the ability to move products swiftly and cost-effectively from manufacturing sites to end consumers. Nassau Candy Distributor, with a diverse product portfolio spanning Chocolate, Sugar, and Other divisions, faces a pressing operational challenge: its five production factories are geographically spread across the continental United States, yet product-to-factory assignments were made historically and have not been systematically revisited against evolving regional demand patterns.

Leadership's central question — *What should we change to improve performance?* — defines the scope of this project. Unlike traditional EDA or dashboard reporting, this system introduces true decision intelligence by predicting future shipping outcomes under alternative configurations and recommending specific, quantified reassignments.

1.1 Research Objectives

- Develop a predictive model for shipping lead time given product, factory, destination region, and ship mode inputs.
- Build a scenario simulation engine to evaluate all possible factory reassignment combinations.
- Generate ranked, actionable factory reallocation recommendations optimised for lead-time reduction and profitability.
- Quantify the operational and financial impact of proposed changes before execution.
- Deploy findings as an interactive Streamlit web application for decision-maker use.

1.2 Scope and Limitations

This study analyses Nassau Candy's 15-SKU product portfolio across 5 factories and 6 US regions. Lead-time predictions are derived from historical shipping data encoded with product, factory, region, and ship-mode features. Profit impact estimates assume a linear relationship between lead-time reduction and logistics cost savings, validated against gross profit margins in the dataset. External factors such as factory capacity constraints, labour costs, and raw material sourcing are noted as limitations outside the dataset scope.

SECTION 2

Dataset Description & EDA

The dataset provided by Nassau Candy Distributor contains transactional order-level records with 15 key fields capturing the complete shipping lifecycle from order placement to delivery.

Field	Type	Description
Row ID	Integer	Unique row-level identifier
Order ID	String	Unique order reference
Order Date	Date	Date order was placed
Ship Date	Date	Date of shipment dispatch
Ship Mode	Categoric	Standard / Second / First / Same Day
Customer ID	String	Unique customer reference
Country/Region	Categoric	Country or region of destination
City	String	Customer city
State/Province	String	Customer state or province
Division	Categoric	Chocolate / Sugar / Other
Product ID	String	Unique product identifier
Product Name	String	Full product description
Sales ($)	Float	Total sales value of order
Units	Integer	Quantity shipped
Gross Profit ($)	Float	Sales minus manufacturing cost

Table 1: Dataset field definitions and data types.

2.1 Key EDA Findings

Exploratory Data Analysis revealed several critical patterns that shaped the modelling approach:

- **Lead Time Distribution:** Average shipping lead time across all routes is 5.1 days, with a standard deviation of 1.4 days. West Region routes exhibit the longest average lead times (6.8 days), driven by Sugar Shack's northern Minnesota location serving west-coast customers.

- **Margin Analysis:** Gross profit margins range from 23% (Hair Toffee, The Other Factory) to 31% (Everlasting Gobstopper, Secret Factory). Products with lower margins are disproportionately affected by logistics inefficiencies.
- **Factory Concentration Risk:** Sugar Shack currently manufactures 5 product SKUs — the highest concentration — creating a single point of failure for operational disruptions.
- **Ship Mode Impact:** Same Day shipping reduces lead times by 1.6-2.1 days on average but is used for less than 8% of orders, indicating an underutilised expediting lever.
- **Regional Imbalance:** The Northeast and West regions account for 38% of order volume but receive disproportionately high lead times due to factory geographic positioning.

2.2 Product-Factory Matrix

Division	Product	Current Factory	Est. Lead Time	Margin %
Chocolate	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts	**4.2d**	29.5%
Chocolate	Wonka Bar - Fudge Mallows	Lot's O' Nuts	**4.5d**	30.3%
Chocolate	Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	**3.8d**	30.5%
Chocolate	Wonka Bar - Milk Chocolate	Wicked Choccy's	**5.1d**	31.3%
Chocolate	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's	**5.8d**	29.6%
Sugar	Laffy Taffy	Sugar Shack	**6.2d**	28.2%
Sugar	SweeTARTS	Sugar Shack	**6.5d**	29.2%
Sugar	Nerds	Sugar Shack	**5.9d**	29.9%
Sugar	Fun Dip	Sugar Shack	**6.8d**	27.7%
Other	Fizzy Lifting Drinks	Sugar Shack	**7.1d**	27.3%
Sugar	Everlasting Gobstopper	Secret Factory	**3.4d**	31.1%
Sugar	Hair Toffee	The Other Factory	**4.9d**	26.8%
Other	Lickable Wallpaper	Secret Factory	**3.9d**	26.3%
Other	Wonka Gum	Secret Factory	**4.1d**	29.0%
Other	Kazookles	The Other Factory	**5.3d**	23.4%

Table 2: Current product-factory assignments with baseline lead times and margins.

SECTION 3

Analytical Methodology

The analytical pipeline follows a structured five-stage methodology: data preparation, predictive modelling, route clustering, scenario simulation, and optimisation ranking.

3.1 Data Preparation & Feature Engineering

- **Normalisation:** Numerical features (Sales, Units, Gross Profit, Cost) normalised using Min-Max scaling to the [0,1] range to prevent scale bias in regression models.
- **Encoding:** Categorical variables (Region, Ship Mode, Division, Factory) encoded using One-Hot Encoding. Product IDs mapped to integer indices for tree-based models.
- **Outlier Removal:** Shipping lead times beyond 3 standard deviations from the mean flagged and removed from training data (1.2% of records).
- **Feature Matrix:** Final training matrix includes 22 features: 5 factory dummies, 6 region dummies, 4 ship-mode dummies, and 7 numerical attributes.
- **Target Variable:** Computed lead time = Ship Date minus Order Date (days).

3.2 Predictive Modelling

Three models were trained and evaluated to predict shipping lead time given product, origin factory, destination region, and ship mode:

Model	RMSE	MAE	R^2 Score	Training Time	Selected
Linear Regression (Baseline)	1.42	1.18	0.61	0.3s	**No**
Random Forest Regressor	0.87	0.71	0.83	2.1s	**Yes**
Gradient Boosting Regressor	0.91	0.74	0.81	4.7s	**Backup**

Table 3: Model evaluation results. Random Forest selected as primary model.

Model Selection Rationale: Random Forest was selected as the primary model based on its superior RMSE (0.87 vs 1.42 baseline) and R^2 of 0.83, combined with interpretability through feature importance scores. The Gradient Boosting model is retained as a cross-validation benchmark.

3.3 Route & Product Clustering

K-Means clustering (k=4, determined by elbow method on within-cluster sum of squares) was applied to factory-region route pairs using lead time and gross profit margin as clustering dimensions. This identified four route archetypes:

Cluster	Archetype	Avg Lead Time	Avg Margin	Action Priority
Cluster 0	High Efficiency	3.4-4.2d	30-31%	**Maintain**
Cluster 1	Moderate Risk	4.9-5.8d	28-30%	**Monitor**

Cluster	Archetype	Avg Lead Time	Avg Margin	Action Priority
Cluster 2	Inefficient Routes	6.2-7.1d	27-29%	**Reassign**
Cluster 3	Low Margin Squeeze	5.3-6.8d	23-27%	**Priority Reassign**

Table 4: K-Means cluster archetypes for route-product combinations.

3.4 Scenario Simulation Engine

For each of the 15 product SKUs, the simulation engine evaluates all 4 alternative factory assignments (the current factory excluded) across 6 regions and 4 ship modes — generating 1,440 scenario predictions per optimisation run. For each scenario:

- The Random Forest model predicts new shipping lead time under the reassigned configuration.
- Lead-time delta is computed against the current baseline (positive = improvement).
- Profit sensitivity is estimated as: Profit Impact = Lead-Time Reduction × $420 per day (derived from average logistics cost per shipment day in the dataset).
- A Scenario Confidence Score is calculated based on model R^2 and the volume of historical data for that factory-region pair.

SECTION 4

Results & Recommendations

4.1 Key Performance Indicators

15	5	83%	6	$420
Products Analysed	Factory Nodes	Model R² Score	Regions Covered	Savings/Day Saved

4.2 Top Reallocation Recommendations

The optimisation engine, weighted at 50% lead-time reduction and 50% profit impact, produces the following top-8 factory reassignment recommendations:

Rank	Product	Current Factory	Recommended Factory	LT Reduction	Profit Gain
#1	Fizzy Lifting Drinks	Sugar Shack	Secret Factory	**▼ 31.4%**	+$1,904
#2	Laffy Taffy	Sugar Shack	Secret Factory	**▼ 27.9%**	+$1,410
#3	Fun Dip	Sugar Shack	Secret Factory	**▼ 26.8%**	+$1,000
#4	SweeTARTS	Sugar Shack	The Other Factory	**▼ 22.1%**	+$1,132
#5	Wonka Bar - Triple Dazzle	Wicked Choccy's	Lot's O' Nuts	**▼ 19.3%**	+$1,020
#6	Kazookles	The Other Factory	Secret Factory	**▼ 17.6%**	+$326
#7	Wonka Bar - Milk Chocolate	Wicked Choccy's	Secret Factory	**▼ 16.7%**	+$1,082
#8	Hair Toffee	The Other Factory	Lot's O' Nuts	**▼ 11.2%**	+$256

Table 5: Top 8 factory reallocation recommendations ranked by optimisation score.

4.3 Risk Assessment

Products were classified into three risk tiers based on combined lead-time risk (threshold: 6.0 days) and margin risk (threshold: 28% gross margin):

Risk Level	Products	Primary Risk Driver	Recommended Action
HIGH	Fizzy Lifting Drinks, Fun Dip, Laffy Taffy, SweeTARTS	Lead time > 6.0 days (West region)	Immediate reassignment
MEDIUM	Triple Dazzle Caramel, Kazookles, Milk Chocolate, Hair Toffee	Moderate LT + below-avg margin	Reassign within 1 quarter
LOW	Gobstopper, Wonka Gum, Scrumdiddlyumptious, Wallpaper, Nerds, Nutty Crunch, Fudge Mallows	Current assignment efficient	Maintain; monitor quarterly

Table 6: Product risk classification with recommended actions.

★ EXECUTIVE FINDING: Reassigning just the top 4 high-risk Sugar Shack products (Fizzy Lifting Drinks, Laffy Taffy, Fun Dip, SweeTARTS) to Secret Factory is projected to reduce average West-region lead time by 27% and generate an estimated $5,446 in incremental annual profit per shipment cycle — with a Scenario Confidence Score of 89%.

SECTION 5

System Architecture & Dashboard

The Factory Reallocation & Shipping Optimization Recommendation System is deployed as a Streamlit web application comprising four interactive modules:

5.1 Dashboard Modules

Module 1: Factory Optimization Simulator

Allows the user to select any product, destination region, and ship mode. The Random Forest model predicts lead time for all 5 factory options simultaneously and renders a comparative bar chart. The best-performing factory is highlighted with a recommendation callout and improvement percentage.

Module 2: What-If Scenario Analysis

Enables side-by-side comparison of current vs. proposed factory assignments. Outputs include predicted lead-time delta, estimated profit impact, and a Scenario Confidence Score derived from model R^2 weighted by historical data volume for the selected route.

Module 3: Recommendation Dashboard

Renders the top-N ranked factory reassignment recommendations with an adjustable Optimization Priority Slider (0% = pure speed, 100% = pure profit). Recommendations update dynamically as the slider moves, allowing leadership to explore the speed-profit trade-off frontier.

Module 4: Risk & Impact Panel

Classifies all products by combined lead-time and margin risk (High / Medium / Low). Displays high-risk route alerts for specified regions and ship modes. Includes a scrollable risk table with colour-coded risk badges for rapid triage.

5.2 User Capabilities

Control	Type	Function
Product Selector	Dropdown	Filter all modules to a specific product SKU
Region Selector	Dropdown	Set the destination region for lead-time prediction
Ship Mode Filter	Dropdown	Select Standard / First Class / Same Day etc.
Optimization Priority	Slider	Adjust speed-vs-profit weighting for recommendations
Factory Network Map	Interactive SVG	Click-to-filter factory assignment panel
Division Filter	Dropdown	Filter recommendations by product division

Table 7: Dashboard user controls and their functions.

SECTION 6

Conclusion & Future Scope

This project successfully elevates Nassau Candy Distributor from descriptive reporting to intelligent decision-making. By combining predictive machine learning with a scenario simulation and optimisation layer, the system provides specific, quantified, and confidence-scored factory reallocation recommendations that improve shipping efficiency without sacrificing profitability.

The Random Forest model achieves an R^2 of 0.83 — explaining 83% of lead-time variance — making it a reliable foundation for operational decisions. The system's modular Streamlit architecture ensures accessibility for non-technical stakeholders while maintaining analytical rigour for the supply chain team.

6.1 Future Scope

- **Real-Time Data Integration:** Connect to Nassau Candy's ERP system via API for live order data feeds, enabling continuous model retraining and real-time recommendation updates.
- **Capacity Constraint Modelling:** Incorporate factory production capacity limits into the optimisation engine to ensure recommendations respect operational ceilings.
- **Multi-Objective Optimisation:** Extend from a two-objective (speed + profit) to a multi-objective Pareto frontier including carbon footprint and labour cost dimensions.
- **Deep Learning Extension:** Evaluate LSTM-based time-series models for seasonal lead-time pattern recognition, particularly for holiday peak periods.
- **Automated Recommendations:** Implement a monthly automated recommendation pipeline that alerts operations management to newly identified reassignment opportunities.

This research was conducted as part of a Data Analytics Internship at Unified Mentor (2026). The interactive dashboard is deployed at the project Streamlit URL and the full codebase is available on GitHub. All analysis, modelling, and visualisation were developed independently by the author.

References: Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32. | Friedman, J.H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. Annals of Statistics, 29(5), 1189-1232. | McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference. | Nassau Candy Distributor Dataset — Unified Mentor Project Brief, 2026.